

Contact Information

- Vitali Liashchuk
- Warsaw, Poland
- +48 571 060 356
- [LinkedIn](#)
- vitali.liashchuk@gmail.com

Summary

Senior Software Engineer with 10+ years of modern C++ experience building cross-platform desktop tools and internal developer tooling for graphics and rendering pipelines (Linux, Windows, macOS). Expertise in Dear ImGui, graphics fundamentals (Vulkan/OpenGL/Metal), and SDK architecture.

Skills

- **Core Technical:** C++ (10+ years, C++11/14/17/20, STL, multithreading, coroutines) | Linux development | Memory management & performance optimization | Concurrency | Cross-platform (Linux, Windows, macOS, iOS, Android)
- **Desktop Tooling:** Dear ImGui | Qt (personal projects) | Internal developer tools | Graphics (Vulkan, OpenGL, Metal)
- **Leadership & Infrastructure:** Technical leadership & mentoring | Architecture design | CI/CD (Jenkins) | Build systems (CMake, Make, Gradle)
- **Game Engine Integration:** Unreal Engine (UE4/UE5) — C++ plugin development, render pipeline integration, UObject/UTexture lifecycle | Unity3D native plugin | Cocos2d-x
- **AI & Tools:** OpenAI API, Retrieval-Augmented Generation (RAG), LlamaIndex | Python, Bash | Emscripten

Professional Experience

Senior Software Engineer (C++), Adverty (Dec 2021–Present)

1. [AdvertySDK](#)

- **Description:** High-performance C++ SDK for developers, enabling cross-platform deployment across mobile, desktop, and web
- **Used intensively:** C++, Metal, Vulkan, OpenGL, Linux/Windows/macOS/iOS/Android
- **Achievements:**
 - Successfully led the development and release of the SDK from scratch for cross-platform deployment (Linux, Windows, macOS, iOS, Android)
 - Led and mentored a team of 2-4 engineers, establishing coding standards and conducting code reviews
 - Introduced TDD processes to the team and mentored engineers on best practices
 - Championed adoption of modern C++20 features including coroutines, improving async code performance and readability
 - Led the SDK architecture design with focus on performance-critical systems
 - Implemented CMake-based build infrastructure with CI/CD integration (Jenkins, GitHub Actions)
 - Designed the SDK API for optimal memory efficiency and cross-platform compatibility
 - Built unit and integration test infrastructure
 - Implemented high-performance graphics library with Vulkan/OpenGL/Metal/DX11 support, focusing on memory efficiency and cache optimization
 - Developed UE4/UE5 C++ plugin (.uplugin, Build.cs module) integrating SDK into Unreal's render pipeline — synchronizes ad frame data to native UE texture objects, managing render thread lifecycle and cross-thread resource ownership
 - Developed Unity3D native plugin integration with equivalent render pipeline hooks
 - Created video player with FFmpeg backend, implementing efficient memory management for multimedia streaming
 - Implemented memory pool allocators and cache optimization strategies for graphics rendering
 - Built a cross-platform in-SDK debug tool using Dear ImGui, enabling real-time visualization of rendering state, coordinate transforms, and SDK internals across Windows and macOS

- Conducted performance profiling and optimization across multiple platforms (Linux, Windows, macOS, iOS, Android)
- Optimized memory footprint and CPU performance for resource-constrained mobile devices

Senior Software Engineer (C++), King.com (Apr 2020–Dec 2021)

1. [Pet Rescue Saga](#)

- **Used intensively:** C++, multithreading, performance optimization
- **Achievements:**
 - Architected high-performance game engine using ECS and data-oriented design, optimizing cache locality and memory access patterns
 - Improved runtime performance through memory layout optimization and cache-friendly data structures
 - Created debugging tools using the ImGui library
 - Led codebase refactoring efforts, improving code maintainability and performance across the codebase

Senior Software Engineer (C++), Vizion Games (Jun 2019–Apr 2020)

1. [Mahjong Treasure Quest](#)

- **Used intensively:** C++, Cocos2d-x, cross-platform development
- **Achievements:**
 - Developed memory-efficient UI system with custom CSS parser for cross-platform rendering
 - Implemented object transformations (transform/scale/rotate) with performance in mind
 - Significantly improved developer tools using ImGui, reducing debugging time and enhancing team productivity
 - Designed and implemented a new Flow Actions System with modular architecture

Senior Software Engineer (C++/C#), Gambino (2015–2019)

1. [Slototron.com](#)

- **Used intensively:** C++, Emscripten, JavaScript, multithreading, request handling
- **Achievements:**
 - Developed secure payment system with C++/JavaScript interop, demonstrating cross-language communication skills
 - Implemented request retry mechanism with concurrency handling, significantly improving client-server reliability
 - Optimized application performance through request batching, memory caching, and resource management strategies
 - Built a runnable prototype using C++/Emscripten for Instant Games, demonstrating cross-platform capability

2. [Gambino Slots](#)

- **Used intensively:** C#, Xamarin, Dart
- **Achievements:**
 - Worked on a C# to Dart converter based on the Roslyn compiler
 - Rewrote the game in Dart from scratch
 - Ported the in-house C# game engine to macOS
 - Integrated OpenGL's scissor mechanism
 - Integrated Firebase push messaging
 - Refactored and modularized the codebase per platform

Earlier Software Engineering Experience (2013–2016)

- Multiple roles at MuGames, Gambino, and Playtika focused on mobile gaming development using C#, Xamarin, and various in-house game engines